



A PROJECT REPORT

ON

Transparency of Ethical Product's and its source

in Supply Chain

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TECHNOLOGY)**

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CERTIFICATE

This is to certify that the project report entitles “**Transparency of Ethical Product’s and its source
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ABSTRACT

As there is an enormous progress in the Information Technology field, there is also a rise in the economic which in turn creates a competition between the companies. In the Blockchain era we have entered the Supply Chain Management System. Compared to the traditional Supply Chain Management System the Supply Chain Management using Blockchain is more effective and useful in its own ways, it also surpasses the limitations of the traditional supply chain management. Supply chain acts as a primary concern in the business field while transporting of the products among the producer and consumer in the right and systematic way. Many companies across the globe face many challenges to manage the supply chain from source to destiny. We have created a system which maintains a transparency of products and its source in Supply Chain using Blockchain. The objective of this system is the development of an integrated information system in which the digital stream of data generated in the supply chain of products can be assigned to these products. That's the reason why the majority of companies nowadays approach this concept. There are significant advantages of adopting Blockchain in supply chain management, they are: Customer Fulfilment, Anti-Counterfeit, Identify the problems, cost efficiency and rise in Economy.

Key Words: Blockchain, Supply chain, source, transparency, products, Counterfeit product, QR code

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LIST OF ABBREVIATIONS**ABBREVIATIONS**

UML

DFD

ILLUSTRATIONS

Unified Modeling Language

Data Flow Diagram

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO PROJECT

Blockchain is a new type of data system that maintains and records all the details regarding to the field in a way that permits multiple stakeholders to confidently share access to the same data and information.

A blockchain is a type of distributed ledger technology (DLT), which means it is a data ledger that is shared by multiple entities operating on a distributed network. The Blockchain technology operates by recording and storing every transaction across the network in a cryptographically linked block structure that is replicated across network participants. Each block in the blockchain has a hash, which is the output of an algorithm that turns the contents of the block into a random mix of letters and numbers. By mathematically validating that the hashes match the expected values, users can trust that the data has not been tampered with. Relevant actors or organizations participating in the distributed network can serve as “nodes” to participate in consensus, a process that keeps each blockchain node in sync and handles the addition of new blocks. There are different mechanisms that could be used to establish consensus that have trade-offs between confidentiality, throughput, and security, and vary depending on the implementation and use case.

There are different types of Blockchains:

- Public Blockchain,
- Private Blockchain
- Consortium Blockchains
- Hybrid blockchains

Depending on the type of blockchain platform being used, blockchains can be designed to provide different levels of access to the data on the blockchain (“on-chain” data). This means that it can provide an increase in transparency of the data between the consumer, while upholding privacy where needed. For instance, a blockchain could enable patients to regulate their own health data and choose who can access their health data. Using blockchain during this way establishes trust in the validity of the information and improves the potential to share the info across silos while keeping sensitive data protected.

They use labelled datasets to train and to render it accurate by changing the parameters of the learning rate. After that, they apply parameters of learning rate to the dataset, the techniques that implement supervised learning such as multilayer perceptron (MLP) to build the model based on the history of the database.

1.2 MOTIVATION BEHIND PROJECT TOPIC

In complex production, a large amount of data is generated which can be the source for value-add for actors in the value chain. Blockchain-based systems provide the potential to capture information along the production chain, to provide means for collaboration and they ensure integrity of information.

The worldwide improvement of an item or innovation consistently accompanies hazard factors, for example, forging and duplication also, use of harmful products. Due to awareness of cruelty free environment most users prefer to know the source of the products they are using.

Forging items can influence the organization's name and the client's wellbeing. Now-a-days discovery of forged item is the greatest test. Counterfeit and Environment harmful products are causing a significant impact on the organization and the client's wellbeing.

Hence, the creators of products are defying enormous misfortune. India and different nations are battling to create an environment friendly product.

1.3 AIM AND OBJECTIVE(S) OF THE WORK

1.3.1 Project Aim:

To establish a reliable, responsible, modernized, transparent, and secured supply chain against counterfeiting attacks. To ensure integrity of information and to benefit in revenue growth which will help in Economy development.

1.3.2 Project Objectives:

The objectives of this project are:

- The objective is to develop an integrated information system in which the digital stream of data generated in the supply chain of products can be assigned to these products.
- To Design a system using Blockchain to overcome Counterfeit of products.
- To secure the details of the product using a QR code.
- Provide security and trust between the customer and manufacturer by offering data to customer.
- No manipulation of the stored data can be done.
- Furthermore, the information should enable traceability of product, its sub-products and all steps in manufacturing with their associated quality parameters.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION:

Here we present a review on tracing the source of the product in the supply chain. Due to the traditional supply chain Counterfeit products are growing exponentially with the enormous amount of online and black-market. So, there is a strong need to address the challenges of detecting the source of the products and designing appropriate technology to improve detection accuracy. It is one of the active research areas to be explored in the current world.

2.2 RESEARCH PAPERS:

- Design of an efficient image protection method based on QR code (IEEE-2021): In this paper, the author proposed a technique to protect image data based on QR codes and describe the design and verification results. In the aspect of image protection, the technology presented in this paper can enhance security through the creation of a secure image with both de-identification and encryption applied. In terms of QR code, our proposed method optimizes data to meet capacity limits and to eliminate security vulnerabilities.
- A Blockchain-Based Application System for Product Anti-Counterfeiting (IEEE-2021): This paper is the first Blockchain system that proposes a fully functional anti-product forgery system. By paying a very low transaction fee, users of our system no longer need to be concerned about the possibility of acquiring a counterfeited product.
- Fake Product Detection Using Blockchain Technology (IEEE-2021): In this paper, the author proposed a system which is useful for the customer to detect fake products in the supply chain. Customers are able to scan QR codes which is assigned to the product and can get all the information like transaction history, current owner based on which end-user can check whether the products they bought are authentic or not.
- An Efficient Bar/QR Code Recognition System for Consumer Service Applications (IEEE-2020): This paper presents an image processing-based method for automatic analysis of Bar/QR codes. The proposed method involves the filtering of the images which removes the noises associated in the

images. Further, the images are converted into grey scale and binarization is performed. The experiments conducted using this yielded 87% accuracy

- A Blockchain-based Supply Chain Quality Management Framework (IEEE-2020): In this paper, the author proposed a blockchain-based framework. The framework comes up with a theoretical basis for brilliant quality management of the supply chain which is based on the Blockchain Technology. Furthermore, it imparts a foundation to develop theories related to data resource management in distributed, virtual organizations.
- Smart Tags for Brand protection and anti-counterfeiting in the wine industry (IEEE-2020): This paper describes a brand protection and anticounterfeiting solution for the wine industry based on smart tags and Cloud enabled technologies. The main idea utilized in this technique are the usage of smart tags which are used to utilize quick response codes and functional inks supported by the Cloud system and which maintains a two-way communication between the winemaker and the end-user.

2.3 COMPANY'S STANCE TOWARDS BLOCKCHAIN:

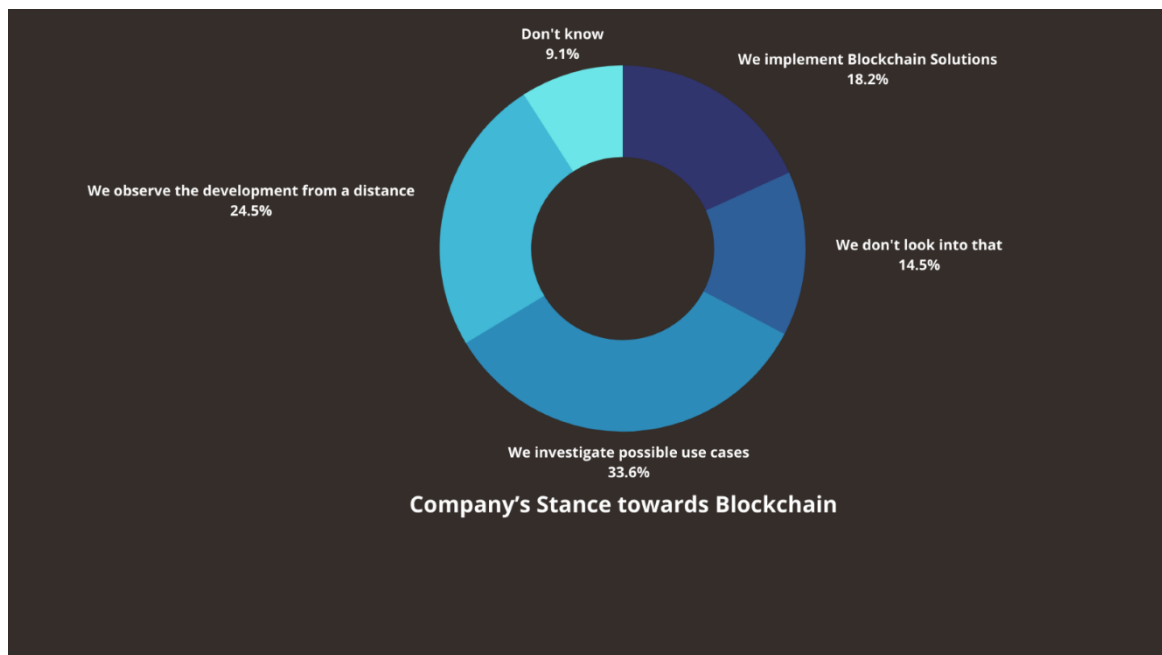


Figure 2.3.1 Company's Stance towards Blockchain

The above pie chart gives us an idea of the company stance towards blockchain technology. Around 24.5% companies observe the development from a distance and do not use the technology. 33.6 % companies investigate the possible use cases. 14.5% companies are not interested in the technology. While 18.2% have successfully implemented the blockchain technology.

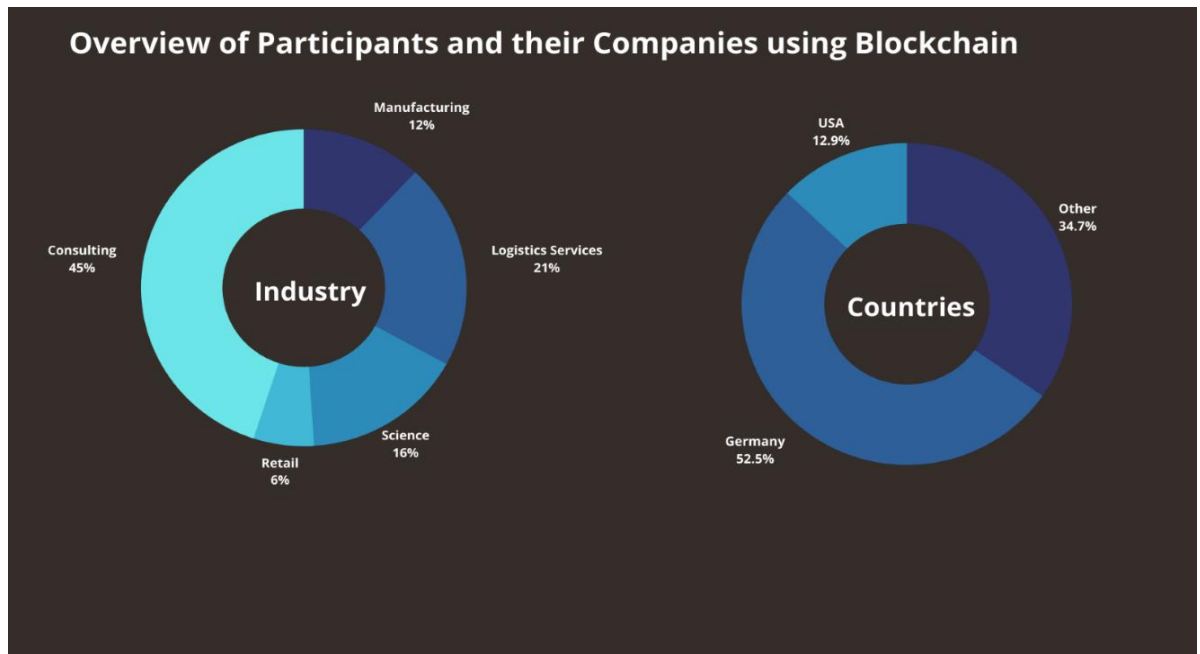


Figure 2.3.2 Overview of Participant's and their Companies using Blockchain

The above pie chart gives the overview of the participants and their companies using blockchain technology.

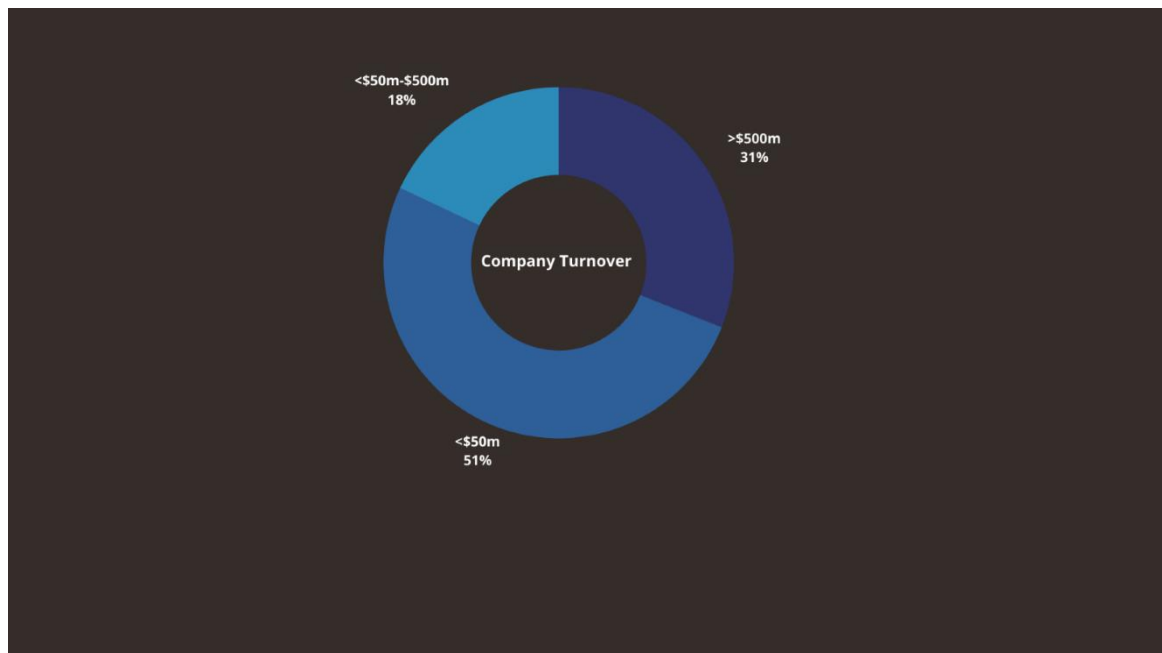


Figure 2.3.3 Overview of Companies turnover using Blockchain

The above pie chart gives the overview of the companies turn over using block chain technology. This pie chart shows us that Blockchain is a major contributor in majority of the companies with a turn over of above 550 million.

CHAPTER 3

PROBLEM DEFINITION AND SCOPE

3.1 PROBLEM DEFINITION

To design and developed a system for supply chain management using blockchain technology, in this work system carried out three different modules like supplier, vendor and admin as well. Each transaction has stored into the blockchain which eliminate all network data attacks in P2P environment for health care systems.

3.2 SCOPE OF THE PROJECT

- ▶ Blockchain, the digital ledger technology that can securely maintain continuously growing lists of data records and transactions, has the power to potentially transform medicine and health care supply chain management, according to industry experts.
- ▶ By simplifying and expediting the way the transaction industry processes data in such areas as revenue cycle management, health data interoperability and supply chain validation, blockchain has the power to dramatically reduce back-office data input and maintenance costs and improve data accuracy and security.

CHAPTER 4

PROJECT PLAN

4.1 TIME LINE OF THE PROJECT:

The development of the project started with two phases in mind. The first phase consists of research across various domain to select the topic which will be beneficial to the society. The research part of the project after the selection of topic took around 2-3 months. The second phase is the implementation phase which is still in process. The complete implementation of the project will be done in upcoming 3-4 months.

4.2 FEASIBILITY STUDY:

In figure 1.1, 1.2, 1.3 which states the overview of participants and companies those who use block chain technology. The companies using it experiences a huge turnover. This pie chart shows us that Blockchain is a major contributor in majority of the companies with a turnover of above 550 million.

4.3 RISK ANALYSIS:

Blockchain is an emerging technology which is still in the proof-of-concept phase, there are also some of blockchain types who have not yet proven themselves in an industrial environment consequently real threats and challenges will not appear in the few newt years of its implementation. This implementation must be carefully analysed and followed before being adopted because any little mistake in implementation would have major risks and bad consequences.

4.4 EFFORT:

The functionalities and architecture of the prototype were designed and developed in an iterative process. Several workshops with 2 - 4 persons were held and aspects of different architectural approaches were discussed. Duration for the development of the project took around 11 months.

4.5 NUMBER OF PEOPLE:

- Vaishnavi Pawar, BIT52
- Sumedha Bhagwat, BIT54
- Rutuja Awade, BIT55
- Priya Yadav, BIT57

CHAPTER 5

PROJECT REQUIREMENT SPECIFICATIONS

5.1 SOFTWARE REQUIREMENTS

- Operating System: -Windows XP/7/8
- Programming Language: JAVA/J2EE/
- Tools: Eclipse or Higher, Heidi SQL, JDK 1.7 or Higher
- Database: MySQL 5.1
- MySQL 5.1

5.2 HARDWARE REQUIREMENTS

- Processor: - Intel Pentium 4 or above
- Memory: - 2 GB or above
- Other peripheral: - Printer
- Hard Disk: - 500gb

5.3 NON-FUNCTIONAL INTERFACES

5.3.1 Performance requirements

The performance of the functions and every module must be well. The overall performance of the software will enable the users to work efficiently. Performance of detect face should be fast. Performance of the providing virtual environment should be fast.

5.3.2 Safety requirements

The application is designed in modules where errors can be detected. This makes it easier to install and update new functionality if required.

5.3.3 Software Quality Attributes

Our software has many qualities attribute that are given below: -

- **Adaptability:** This software is adaptable by all users.
- **Availability:** This software is freely available to all users. The availability of the software is easy for everyone.
- **Maintainability:** After the deployment of the project if any error occurs then it can be easily maintained by the software developer.
- **Reliability:** The performance of the software is better which will increase the reliability of the Software.

CHAPTER 6

ARCHITECTURE OF THE SYSTEM

6.1 ARCHITECTURE:

The below figure represents the Architecture of our system.

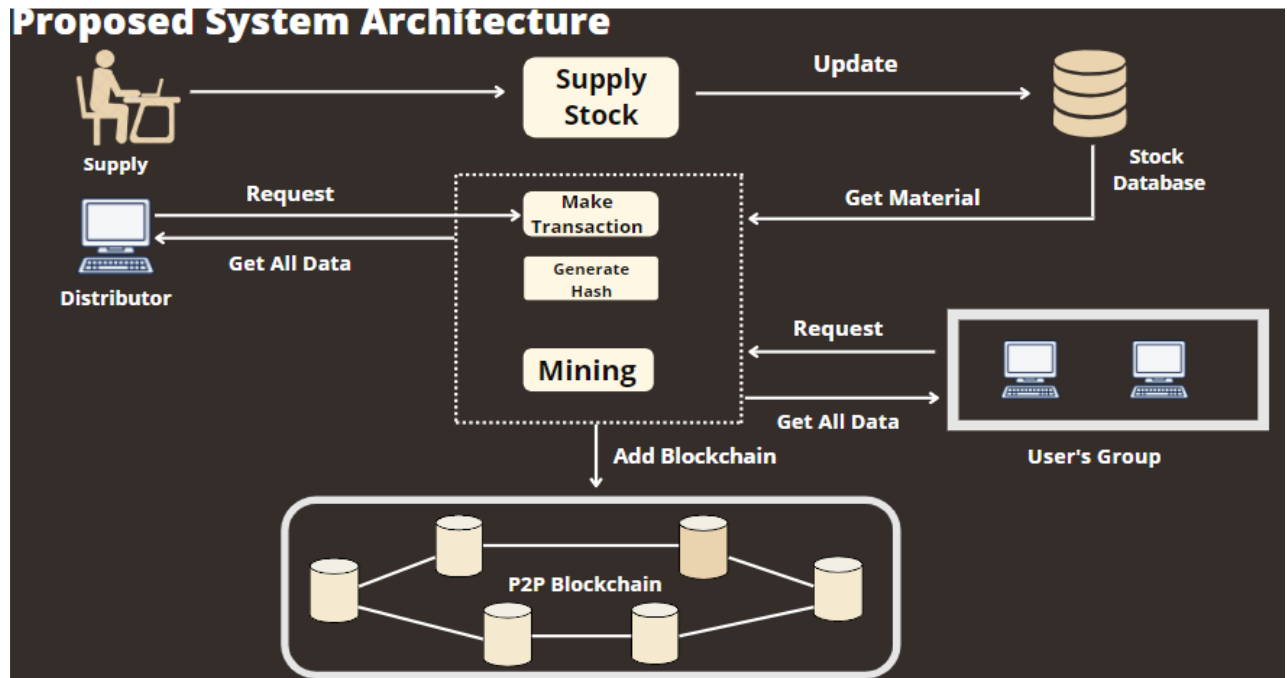


Figure 6.1.1 Architecture of the System

We proposed a system using blockchain technology as a web application for the tracing of products back to its source and for detection of counterfeit products. The proposed system ensures that the products bought are safe and sustainable right from its source which are used in day-to-day life. The proposed system as in the Fig 6.1.1 consists of three main parts, customer or user web application, Manufacturer's or Distributor's web application, and Database.

The first application is the Manufacturers or Distributor's application in which the distributor will have to first register themselves. After registration and login into the application, there will be list of option. The first one is to add a product in which the manufacturer can add the product details. Another option is to show the order in which they can see customers' order details and after that, they can decide the accept or reject the order. The manufacturer also can see whether the product is delivered or not.

This system highlights the implementation of e-transaction using Blockchain for such a proposal from a practical Point of view in both development of the project / deployment of the project and usage contexts. Concluding this work is a potential roadmap for Blockchain technology to be able to support complex applications. Building an electronic transaction system that satisfies the legal requirements of legislators has been a challenge for a long time. Distributed ledger technologies are an exciting technological advancement in the Information Technology world. The proposed system uses Blockchain as a main domain. With the use of Blockchain Supply Chain Management, there will be transparency in supply chain as they will allow the supply chain nodes to verify the authenticity of a product's data. Data plays a crucial role in today's business world. In the traditional SCM the data regarding the product is manipulated and lost in the process so to overcome it a secure encryption and verification technologies are used which are present in Blockchain. This ensures that no data is manipulated and lost. Data stored in the blockchain database is immutable. It is very supportive if we want to keep the integrity of a data and make sure that nobody ever tampers with. There are significant advantages of adopting Blockchain in supply chain management: Customer Fulfilment, Anti-Counterfeit, Identify the problems, cost efficiency and rise in Economy. When we take into account all these factors, we are motivated to generate a system using Blockchain which maintains transparency between the consumer and the manufacturer/ distributor. The same is predicted in the Fig. 6.1.1, which depicts the design.

We proposed a system using blockchain technology as a web application for the tracing of products back to its source and for detection of counterfeit products. The proposed system ensures that the products bought are safe and sustainable right from its source which are used in day-to-day life. The proposed system as in the Fig 4.2 consists of three main parts, customer or user web application, Manufacturer's or Distributor's web application, and Database.

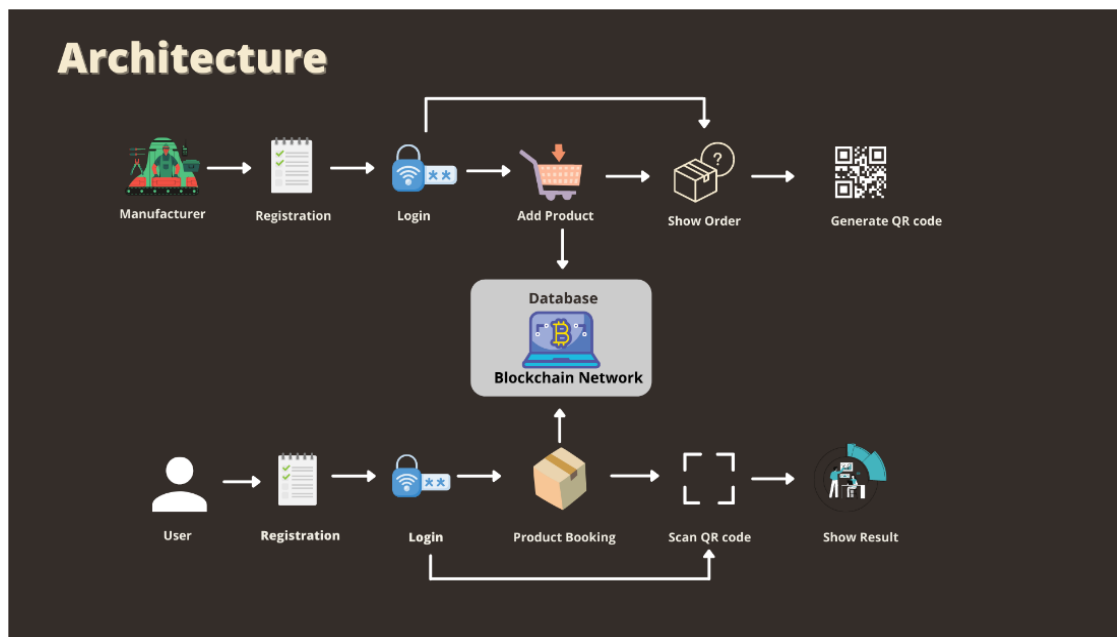


Fig 6.1.2 Architecture of the Application

The first application is the Manufacturers or Distributor's application in which the distributor will have to first register themselves. After registration and login into the application, there will be list of option. The first one is to add a product in which the manufacturer can add the product details. Another option is to show the order in which they can see customers' order details and after that, they can decide the accept or reject the order. The manufacturer also can see whether the product is delivered or not.

A second application is the Customer application in which we have to first register in-app after that we can log in to the application using id and password. In this application, there is an option to show products where customers can see the product details like name, total quantity, price of a product, details of the manufacturer. In that, we can product book the product by inserting the quantity of product.

In this application, we can see orders using show my order where we can see the product details, name, quantity, date, time, price, and status of product which is produced is delivered or not. In this app we have a QR code scanner in which we scan the QR code of the product then it shows the source of the product. There is another option which is a blockchain in which it displays the name of generated block product quantity, generated Hash Value, and the product is corrupted or not.

In this project there is a customer login, in the application. After login, he fills in the details for ordering the product and book the product. The order of the product can be shown to the manufacturer. Manufacturer deciding whether the product request is acceptable or not. After the manufacturer accepting the order of the product it generates the unique QR code of a product. Once an order of product is stored on the network hash code is generated of that product and it is possible

to maintain the transaction of the product. In the proposed system QR code is generated for a particular product. Customers scan the QR code on the product or

package using the smartphone's QR code reader application or customer application have the option of QR code scanner.

After scanning, we get the result of the product source. In the end, the Blockchain system holds these product details along with a history of transactions to enable the tracking of the product along the distribution chain. All the product details, block name, a hash value is stored in the firebase cloud database

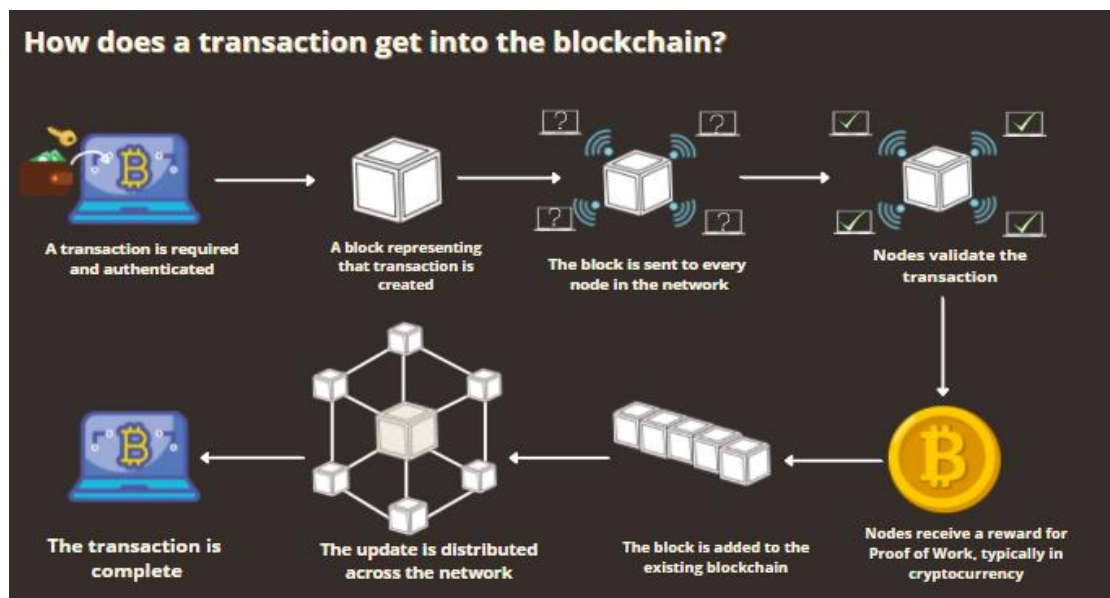


Figure 6.1.3 How does a transaction get into the blockchain

Fig 6.1.3 shows how a transaction gets into the Blockchain. When a transaction is requested and authenticated through the platform a new block is created. The Blockchain technology operates by recording and storing every transaction across the network in a cryptographically linked block structure that is replicated across network participants. The block which is created regarding the transaction is sent to every node in the network. Nodes validate the transaction. Typically, in Cryptocurrency it is noticed that nodes i.e., participants receive a reward for proof of work. Each block in the blockchain has a hash, which is the output of an algorithm that turns the contents of the block into a random mix of letters and numbers.

By mathematically validating that the hashes match the expected values, users can trust that the data has not been tampered with. Relevant actors or organizations participating in the distributed network can serve as “nodes” to participate in consensus, a process that keeps each blockchain node in sync and handles the addition of new blocks. The block is then added to the existing blockchain and the update regarding it is distributed across the network. Hence the transaction is completed.

CHAPTER 7

HIGH LEVEL DESIGN OF THE PROJECT (UML/DFD)

7.1 UML DIAGRAM:

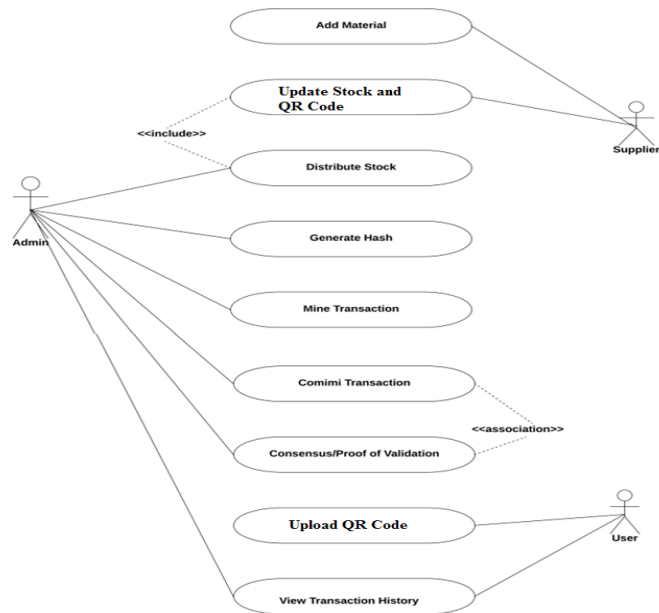


Figure 7.1.1 Use case Diagram

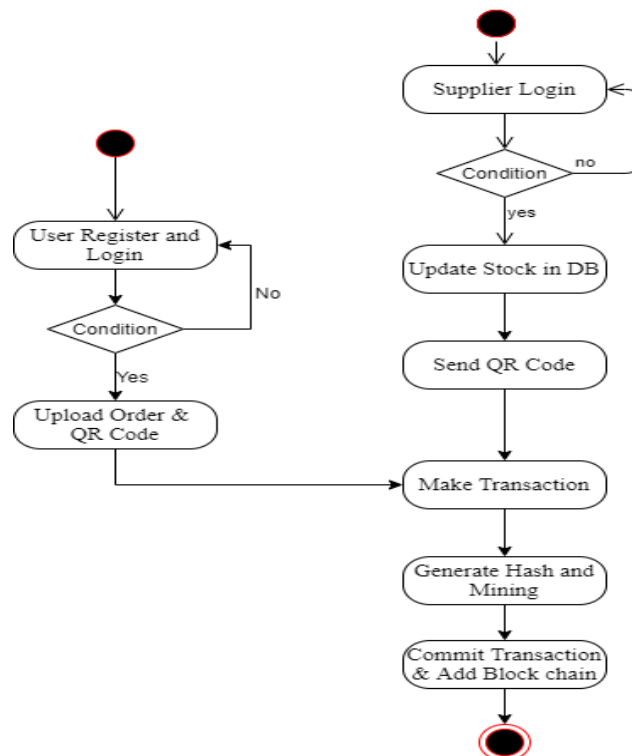


Figure 7.1.2 Activity Diagram

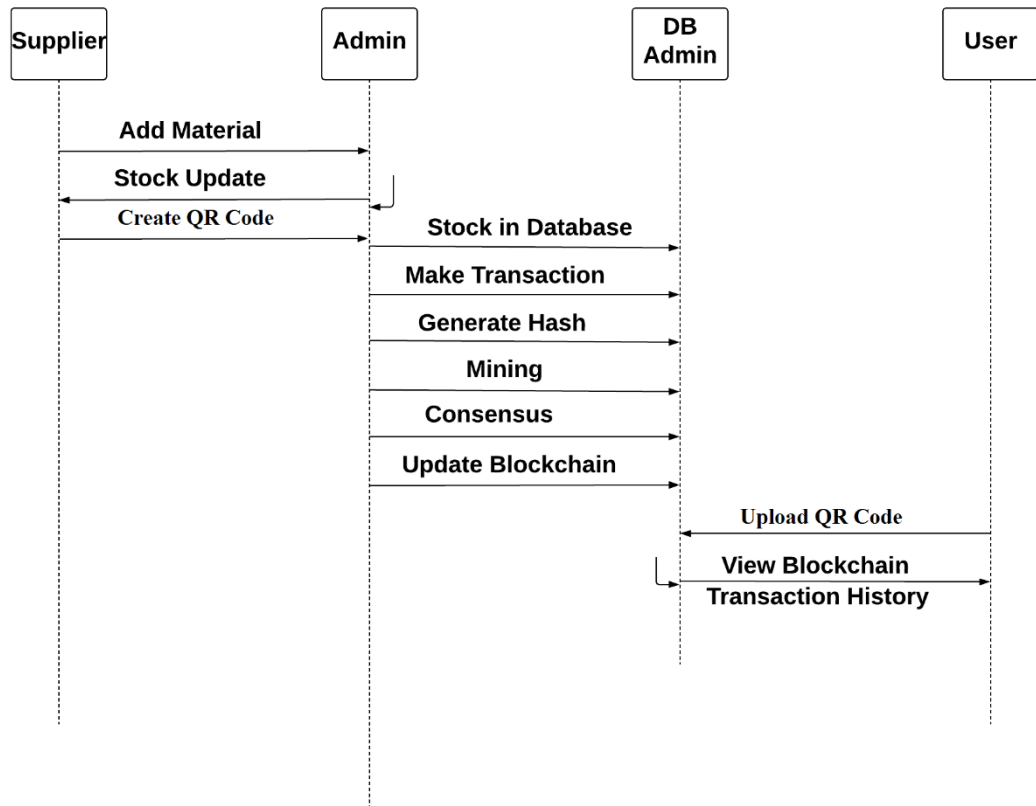


Figure 7.1.3 Sequence Diagram

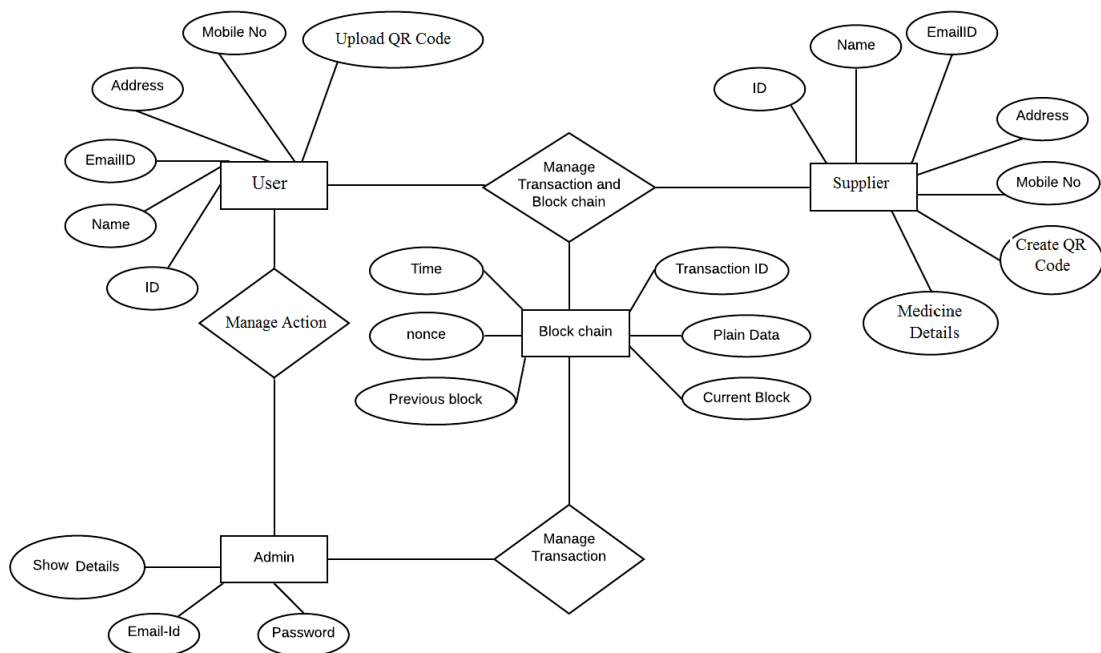


Figure 7.1.4 ER Diagram

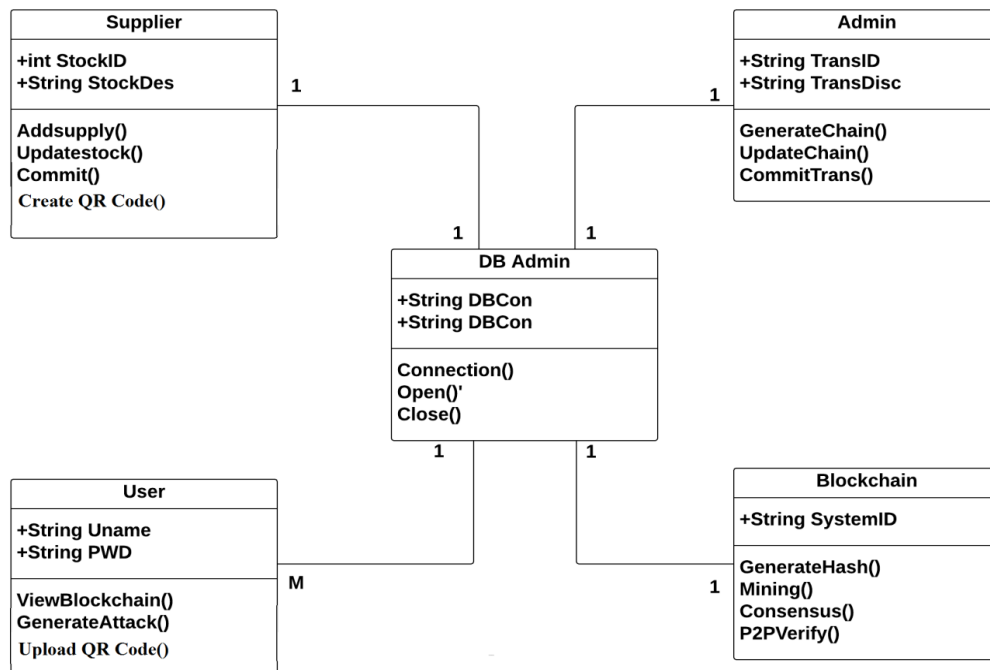


Figure 7.1.5 Class Diagram

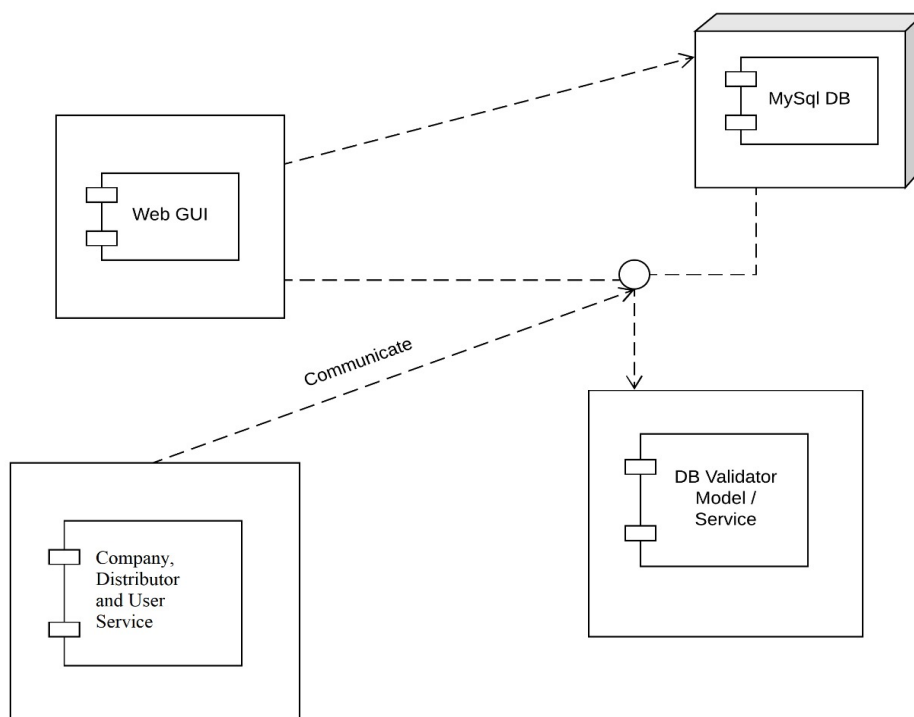


Figure 7.1.6 Component Diagram

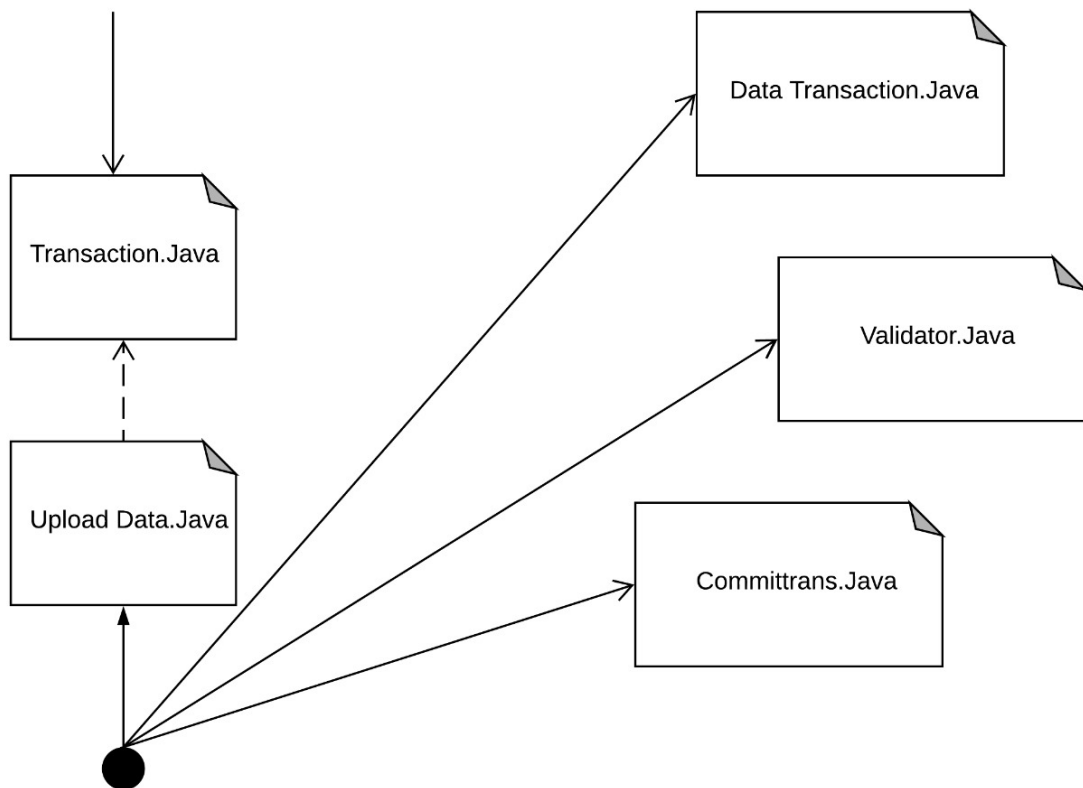


Figure 7.1.7 Deployment Diagram

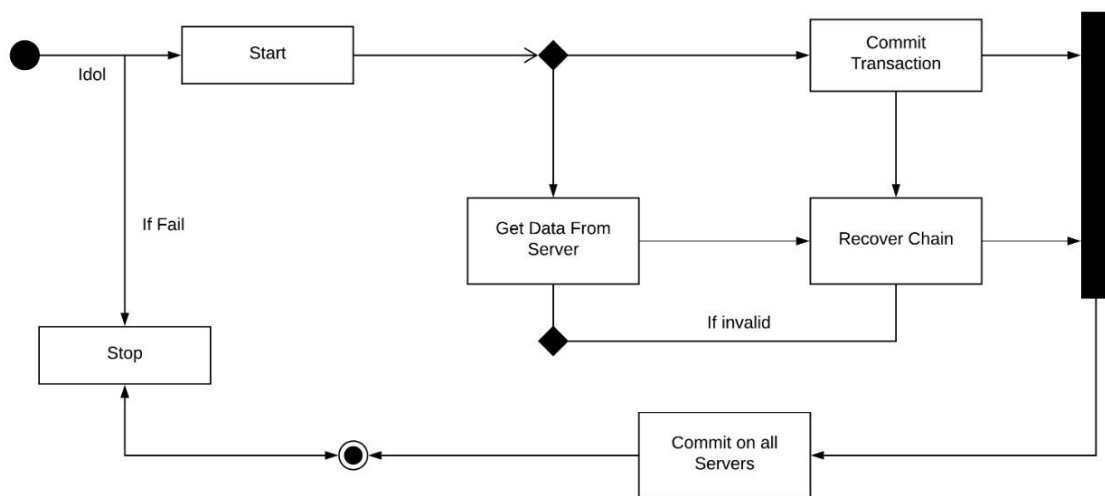


Figure 7.1.8 State machine Diagram

7.2 DFD

7.21. DFD Level 0:

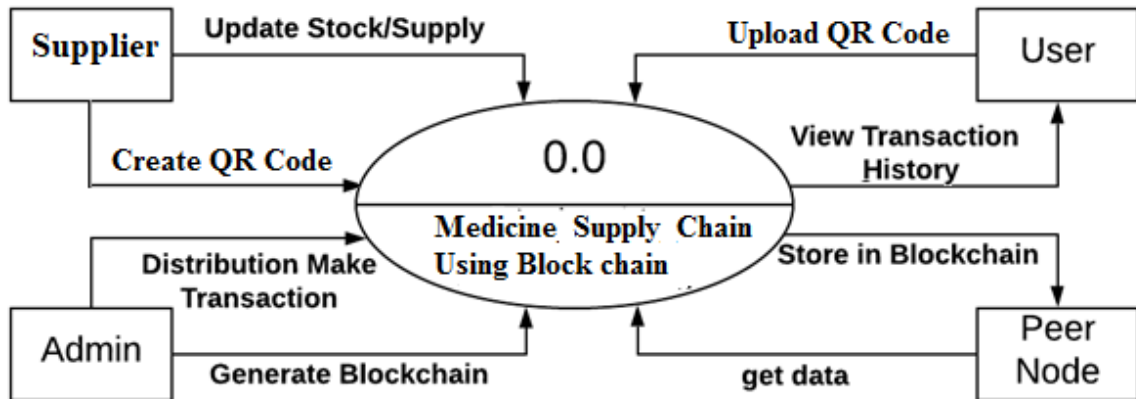


Figure 7.2.1 DFD Level 0

7.2.2 DFD Level Multilevel:

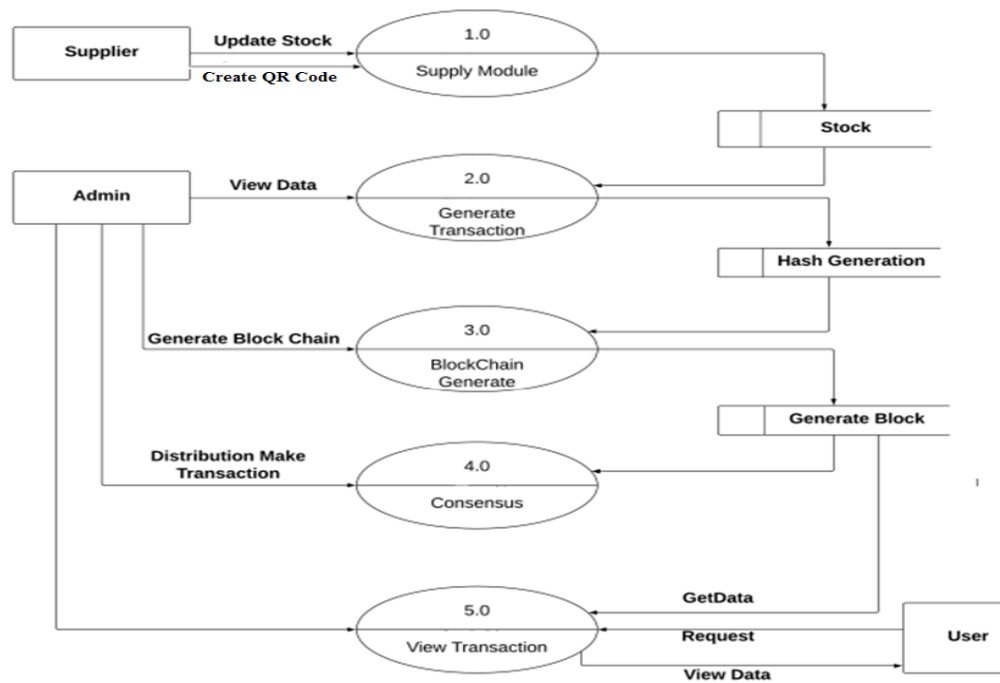


Figure 7.2.2 DFD Multilevel

CHAPTER 8

SYSTEM IMPLEMENTATION

8.1 SHA-256 Algorithm:

SHA 256 may be a part of the SHA 2 family of algorithms, where SHA stands for Secure Hash Algorithm. Published in 2001, it had been a joint effort between the NSA and NIST. The significance of the 256 within the name stands for the ultimate hash digest value, i.e., irrespective of the dimensions of plaintext/cleartext, the hash value will always be 256 bits. The other algorithms within the SHA family are more or less almost like SHA 256. Now, check out knowing a touch more about their guidelines.

8.1.1 Algorithm Working:

Hash Generation: SHA-256

Input: Genesis block, previous hash, data d

Output: According to the given data a hash H will be generated.

- Step 1: Input data as d
- Step 2: Apply SHA 256 from SHA family
- Step 3: CurrentHash= SHA256(d)
- Step 4: ReturnCurrentHash

8.2 PEER-TO-PEER (P2P):

Peer-to-peer (P2P) networks have evolved as a cost-effective and efficient way to distribute videos to a large number of people through the Internet. P2P streaming systems (P2PSS) are difficult to test, analyse, or verify due to their dynamic and heterogeneous character. Complex communication protocols are required for these systems, and these protocols must be verified. Because the central server can soon become overburdened as the number of nodes grows, server-based solutions have a bandwidth scaling difficulty. Peer-to-peer systems are required because they offer a greater number of communication channels and hence scalability. Previously, verification of communication protocols was pursued, especially for security and cryptographic systems. The protocol must be tested to ensure that it does not cause issues like deadlocks or process hunger, and that it has correctness properties like the system gradually achieving specified operating states.

8.2.1 Algorithm: Protocol for Peer Verification

Input: User Transaction query, Current Node Chain CNode[chain], Other Remaining Nodes, Blockchain NodesChain[Nodeid] [chain]

Output: If any chain is invalid it will Recover, else execute the current query

- Step 1: User will generate the transaction using DDL, DML or DCL query
- Step 2: Get current server blockchain Cchain & Cnode[Chain]
- Step 3: For each,
 NodesChain [Nodeid, Chain] $\sum$ (GetChain) n i=1 End for
- Step 4: For each (read i into NodeChain)
 If (!equals NodeChain[i] with (Cchain))
 Flag 1
 Else Continue Commit query
- Step 5: if (Flag == 1) Count = SimilaryNodesBlockchian()
- Step 6: Calculate the majority of server Recover invalid blockchain from specific node
- Step 7: End if
 End for
 End for

8.3 MINING ALGORITHM:

A number of mining algorithms have been developed, each with its own set of strengths and disadvantages. The NSA-created SHA-256 is one of the most widely used algorithms. A mining algorithm is a set of rules used by cryptocurrency miners to determine how to create new blocks.

8.3.1 Algorithm: Mining Algorithm for valid hash creation

Input: Hash Validation Policy P[], Current Hash Values hash_Val

Output: Valid hash

- Step 1: System will generate the hash value (hash_Val) for ith transaction using Algorithm 1.
- Step 2: if (hash_Val.valid with P[])
 Valid hash
 Flag =1

Else

Flag=0

Mine again randomly

- Step 3: Return valid hash when flag=1

8.4 MATHEMATICAL MODEL:

A System has represented by 5-different Phases, each phase works with own dependency System(S).

Here $S = (Q, \Sigma, \delta, q_0, F)$ where,

- Q is a finite set of states.
- Σ is a finite set of symbols which is called the alphabet.
- Δ is the transition function, where $\delta: Q \times \Sigma \rightarrow Q$
- q_0 is the first condition in which any input is processed ($q_0 \in Q$).
- F is a set of final state or states of Q ($F \subseteq Q$)

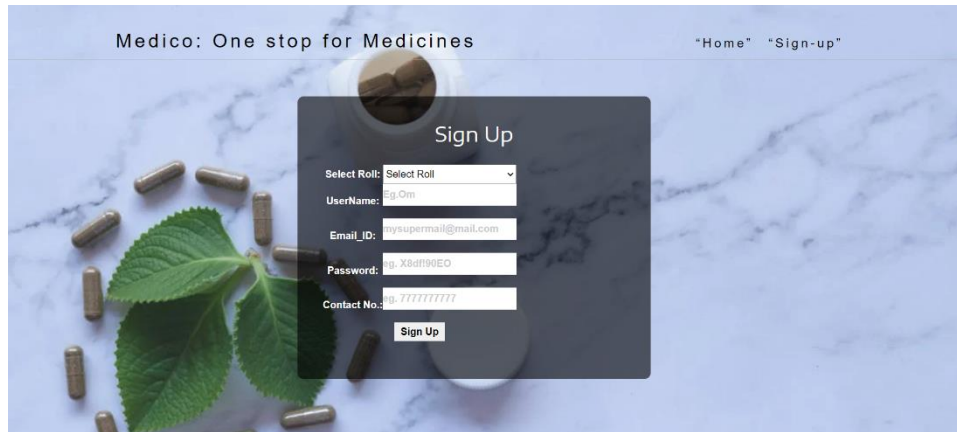
All (n) data nodes will return 1 when each have the same blockchain.

- Q = initial transaction data with genesis block
- $\Sigma = \{\text{SHA-256, Consensus_Val, Mining}\}$
- Δ = Validate all server ($S_1 \subseteq S_2 \subseteq S_3 \subseteq S_4$) all server validation process
- q_0 = Initial transaction $T[0]$
- $F = \{\text{Commit Trans, Get History_Record}\}$
- State $\Rightarrow$ 1: if all chains are validated or same 0: if any $t(n)$ server consists the invalid chain

CHAPTER 9

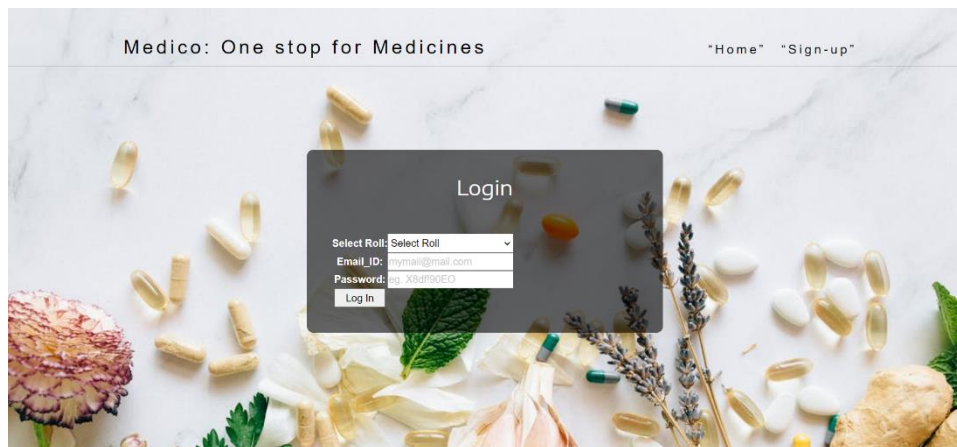
GUI

9.1 GUI OF SYSTEM



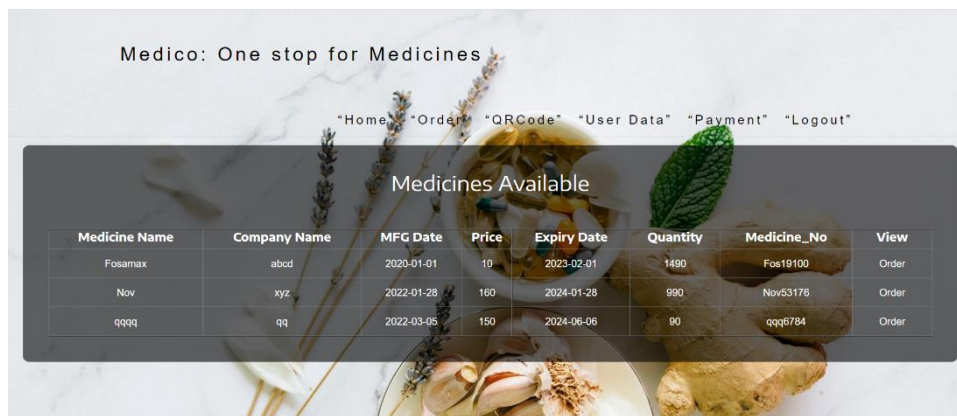
The registration page features a dark grey 'Sign Up' modal box centered on a light blue background with a marble texture and scattered brown capsules. The modal contains the following fields: 'Select Roll' (a dropdown menu), 'UserName' (text input with 'SaOm'), 'Email_ID' (text input with 'mysupermail@mail.com'), 'Password' (text input with 'ng_X&df90EO'), and 'Contact No.' (text input with 'ng_77777777'). A 'Sign Up' button is located at the bottom of the modal. The page header includes the text 'Medico: One stop for Medicines' and navigation links for 'Home' and 'Sign-up'.

Figure 9.1 Registration Page of System



The login page features a dark grey 'Login' modal box centered on a light blue background with a marble texture and scattered yellow and white capsules. The modal contains the following fields: 'Select Roll' (a dropdown menu), 'Email_ID' (text input with 'mymail@mail.com'), and 'Password' (text input with 'ng_X&df90EO'). A 'Log In' button is located at the bottom of the modal. The page header includes the text 'Medico: One stop for Medicines' and navigation links for 'Home' and 'Sign-up'.

Figure 9.2 Login Page of System



The stock page features a dark grey 'Medicines Available' modal box centered on a light blue background with a marble texture and scattered yellow and white capsules. The modal contains a table with the following data:

Medicine Name	Company Name	MFG Date	Price	Expiry Date	Quantity	Medicine_No	View
Fosamax	abcd	2020-01-01	10	2023-02-01	1490	Fos19100	Order
Nov	xyz	2022-01-28	160	2024-01-28	990	Nov53176	Order
qqqq	qq	2022-03-05	150	2024-06-06	90	qqq6784	Order

The page header includes the text 'Medico: One stop for Medicines' and navigation links for 'Home', 'Orders', 'QRCode', 'User Data', 'Payment', and 'Logout'.

Figure 9.3 Stock Of Medicines Available

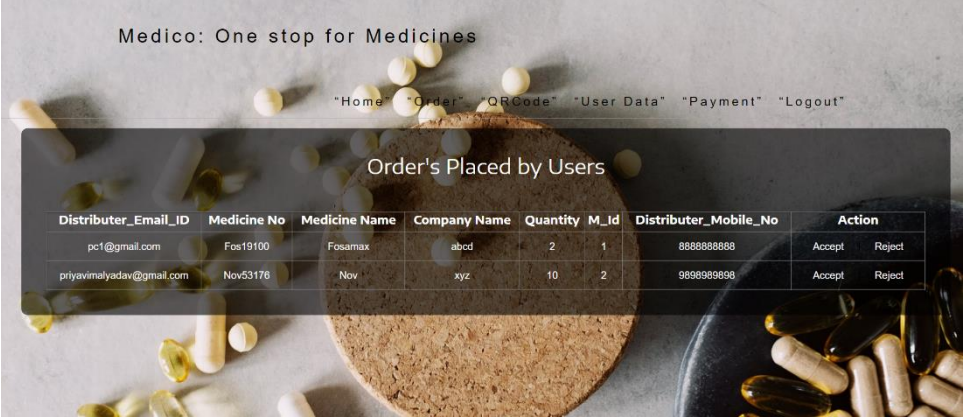


Figure 9.4 Order Placed By User

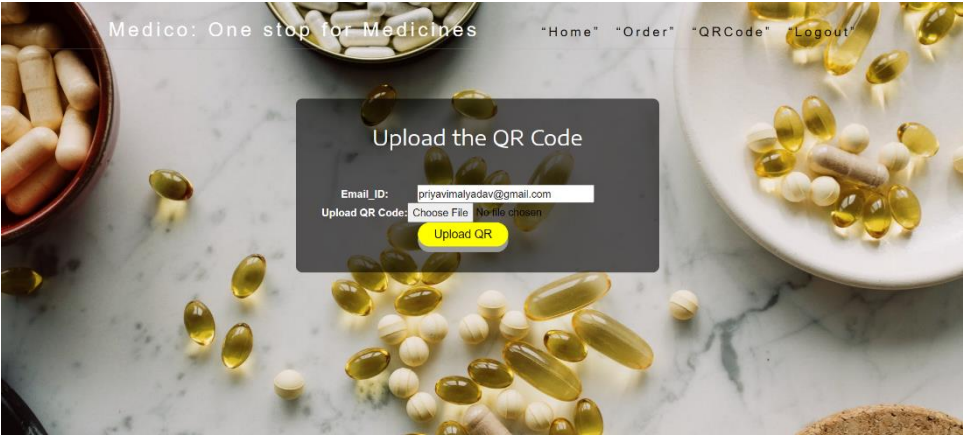


Figure 9.5 QR code upload

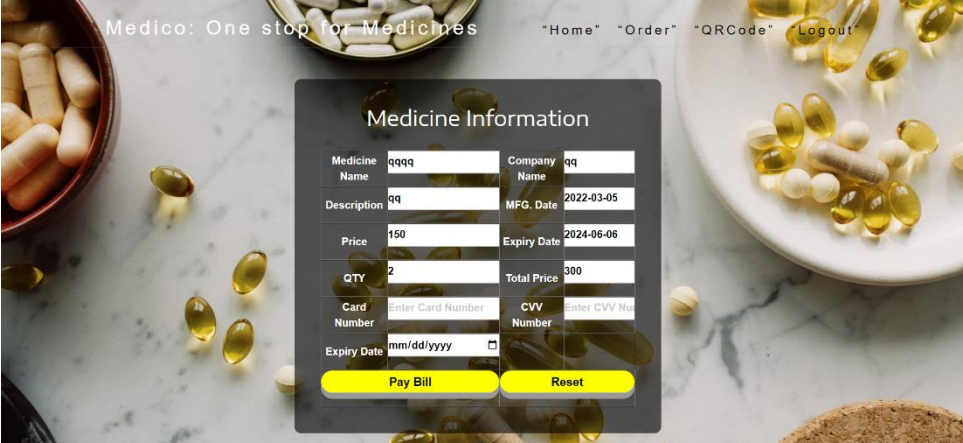


Figure 9.6 User Product Information Page

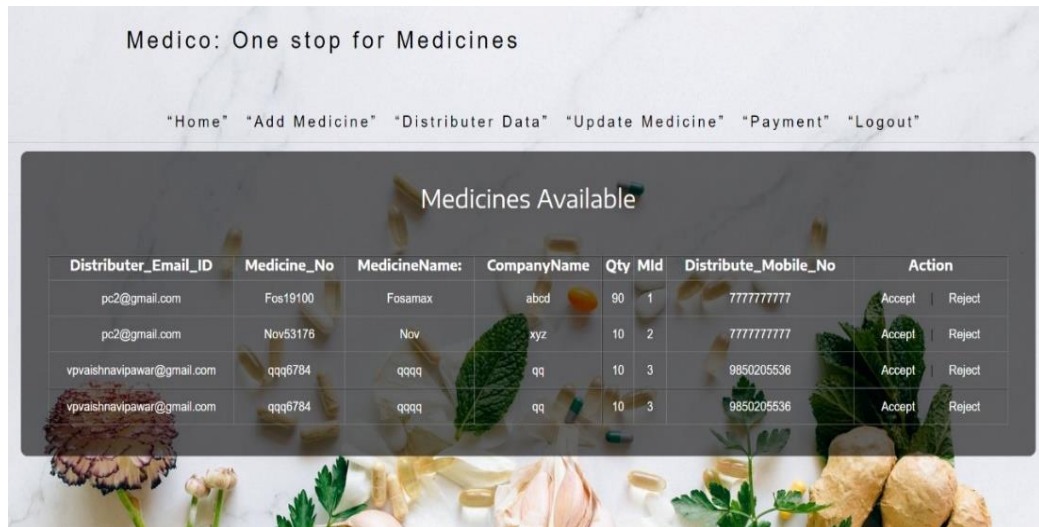


Figure 9.7 Admin Page to Accept or Reject Medicines

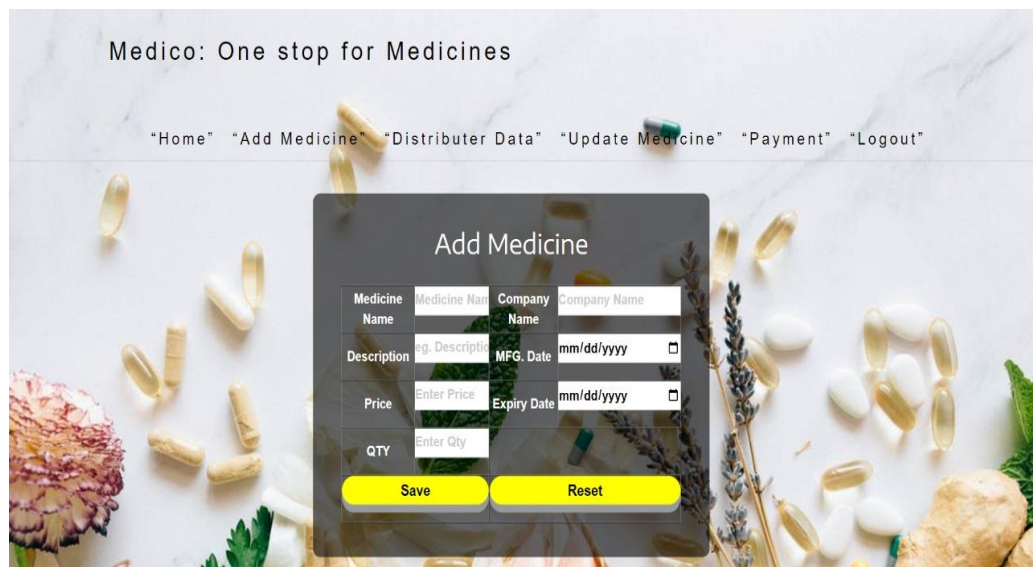


Figure 9.8 Admin Page to Add new Medicines

CHAPTER 10

RESULTS

The time required for each algorithm to validate the Blockchain in four nodes is shown in Figure 10.1. The X axis depicts the algorithm used in the Blockchain, while the Y axis depicts the time needed in milliseconds for each of the four nodes for the given algorithm.



Figure 10.1 Time required for the algorithm in milliseconds

The time required for the consensus algorithm to validate the Blockchain in four nodes is shown in Figure 10.2. The X-axis depicts the algorithm process of the Blockchain for each node, while the Y-axis depicts the time needed in milliseconds for each of the four nodes.

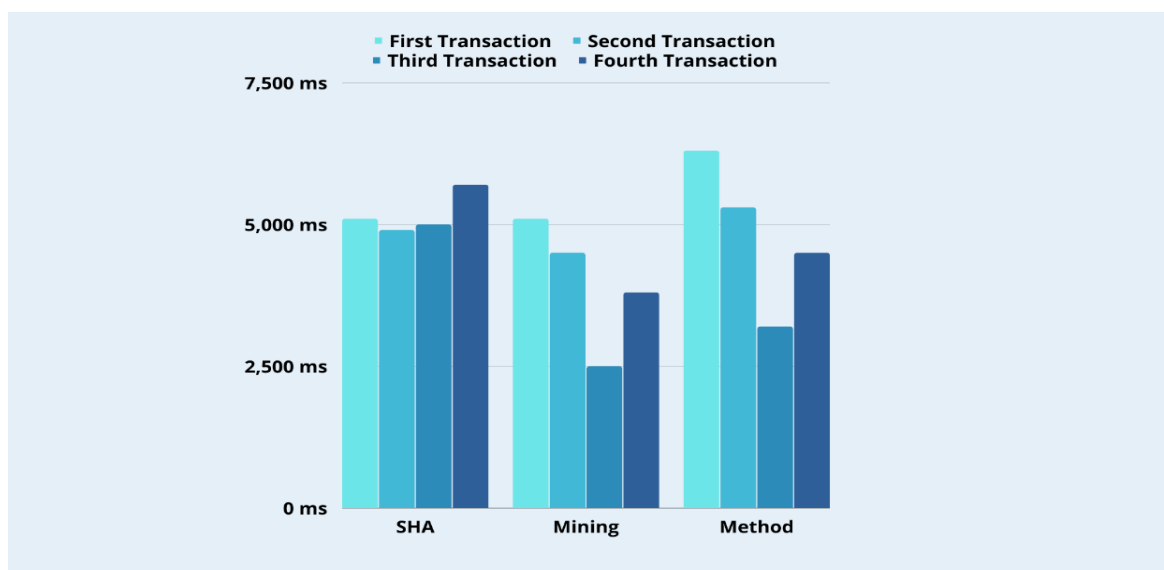


Figure 10.2 Time required to validate the Blockchain in the four nodes



Figure 10.3 QR code generated by the system

The QR code generate is highly encrypted and is unique for each and every product. The QR codes data will be shown only through the application scanner. Due to high level of security and encryption in the Blockchain nodes. The data stored cannot be tampered with, maintaining transparency between the consumer and manufacturer which in turn helps in building the trust between them. The below QR Code in Fig 10.3 is generated by the system after a successful order placement. The QR Code is only scannable only through the system, it won't generate any data after scanning through other system or application. After successful scanning, the system will show the user the details and source of the product.

CHAPTER 11

TESTING

11.1 PURPOSE

Testing is an investigation conducted to provide stakeholders with information about the quality of the product or service under test. Software testing also provides an objective, independent view of the software to allow the business to appreciate and understand the risks of software implementation. Test techniques include, but are not limited to, the process of executing a program or application with the intent of finding software bugs. Software testing can also be stated as the process of validating and verifying that a software program or application or product:

1. Meets the business and technical requirements that guided its design and development;
2. Works as expected; and
3. Can be implemented with the same characteristics

11.2 SCOPE OF TESTING

Software testing, depending on the testing method employed, can be implemented at any time in the development process. However, most of the test effort occurs after the requirements have been defined and the coding process has been completed. As such, the methodology of the test is governed by the software development methodology adopted. Different software development models will focus the test effort at different points in the development process. Newer development models, such as Agile, often employ test driven development and place an increased portion of the testing in the hands of the developer, before it reaches a formal team of testers. In a more traditional model, most of the test execution occurs after the requirements have been defined and the coding process has been completed.

11.3 TEST PLAN

To test this application we are going with proper sequencing of testing like unit, integration, validation, GUI, Low level and High level test cases, major scenarios

likewise. We will go with the GUI testing first and then integration testing. After integration testing performs the high level test cases and major scenarios which can affect the working on the application. We will perform the testing on the data transmitted using the various inputs and outputs and validate the results. It also intends to cover any deviations that the project might take from the initially agreed Test Strategy in terms of scope, testing methodology, tools, etc.. This test plan covers details of testing activities for this project and scope.

11.4 TYPE OF TESTING:

11.4.1 Testing Unit:

Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

11.4.2 Regression Testing:

Regression testing is a software testing practice that ensures an application still functions as expected after any code changes, updates, or improvements. Regression testing is responsible for the overall stability and functionality of the existing features.

11.4.3 Smoke Testing:

Smoke Testing comes into the picture at the time of receiving build software from the development team. The purpose of smoke testing is to determine whether the build software is testable or not. It is done at the time of” building software.” This process is also known as” Day 0”.

It is a time-saving process. It reduces testing time because testing is done only when the key features of the application are not working or if the key bugs are not fixed. The focus of Smoke Testing is on the workflow of the core and primary functions of the application.

11.4.4 System Testing:

System Testing is a type of software testing that is performed on a complete integrated system to evaluate the compliance of the system with the corresponding requirements. In system testing, integration testing passed components are taken as input. System Testing is a black-box testing.

System Testing is performed after the integration testing and before the acceptance testing.

CHAPTER 12

OTHER SPECIFICATION

12.1 ADVANTAGES OF BLOCKCHAIN TECHNOLOGY:

- Producers will have security. They will be able to receive a reasonable percent of profit. They will have knowledge regarding the various factors affecting their crops (i.e., unprecedented seasonal changes, soil quality, water availability, etc.) and thus will be able to take well informed decisions.
- Traders can eliminate competition.
- Processors will be able to procure goods they specifically want for the product they are trying to produce and also control the quality from the bottom up.
- Traceability: Players in different tiers can be held responsible for any damages.
- Increase in multidirectional information transfer.
- Overall increase in efficiency: Only required amount of goods will be produced thus decreasing wastages and increasing profits.
- Using blockchain in this way establishes trust in the validity of the data and improves the ability to share the data across silos while keeping sensitive data protected.
- Blockchain introduces a means to share data across multiple, possibly mistrustful, parties without the necessity for an intermediary, reducing the need to revalidate and reconcile data.

12.2 LIMITATIONS OF BLOCKCHAIN TECHNOLOGY:

- **Lack of Awareness: -**

There is a lot of discussion about blockchain, but people do not know the true value of blockchain and how they could implement it in different situations.

- **Limited availability of technical talent: -**

Today, there are a lot of developers available who can do a lot of different things in every field. But in the blockchain technology, there are not so many developers available who have specialized expertise in blockchain technology.

- **Immutable: -**

In immutable, we cannot make any modifications to any of the records. It is very helpful if you want to keep the integrity of a record and make sure that nobody ever tampers with it. But immutability also has a drawback. We can understand this, in the case, when you want to make any revisions, or want to go back and make any reversals.

- **Key Management: -**

As we know, blockchain is built on cryptography, which implies that there are different keys, such as public keys and private keys. When you are dealing with a private key, then you are also running the risk that somebody may lose access to your private key.

- **Scalability: -**

Blockchain like bitcoin has consensus mechanisms which require every participating node to verify the transaction. It limits the number of transactions a blockchain network can process.

- **Consensus Mechanism: -**

In the blockchain, we know that a block can be created in every 10 minutes. It is because every transaction made must ensure that every block in the blockchain network must reach a common consensus

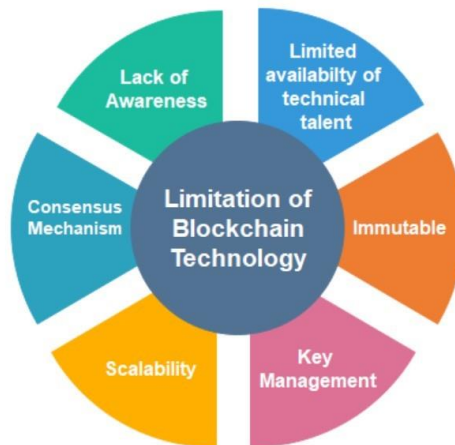


Figure 12.2.1 Limitations of Blockchain Technology

12.3 APPLICATIONS OF BLOCKCHAIN TECHNOLOGY:

Blockchain technology has made significant strides in its development and widespread adoption in recent years, and there's no sign of it slowing down. In fact, according to Deloitte's 2021 Global blockchain development Survey, almost 76% of

executives surveyed said they expect digital assets will be a solid alternative to fiat currencies for global finance in the next 5–10 years.

Below are some applications of Blockchain Technology: -

1.Faster Transactions: - Due to the secured & decentralized nature of blockchain technology, it's incredibly difficult (if not impossible) for hackers or nefarious parties to tamper with transactions. Blockchain-verified data is highly secure and trustworthy, meaning transactions can be processed much faster than in today's world without compromising security.

2.Digital Identity: - Today, we use passwords and authentication questions to prove who we are online. Blockchain could replace this system with a digital identity that is safe, secure, and easy to manage.

3.It'll Be Easier to Treat Health Problems: - As we mentioned earlier, digital identities could be stored and managed through blockchain technology and this goes for health records as well.

As you can see, there's a ton of potential for blockchain technology and it's constantly expanding. Also, the future of blockchain technology looks bright, and when you consider that it's already showing promise in almost every industry, it seems like the best is yet to come.

12.4 FUTURE SCOPE:

We believe that ultimately global trade will evolve into a battle of supply chains, where efficiency, block chain optimization, and innovative financing will yield best in breed organizations. As blockchain gains momentum, "there will be real-time visibility, reduced complexities, improved accuracy and efficiency in the system, which in turn will help reduce costs and enhance trust.

Its usefulness will not be limited to shippers or customers but will extend to lowering trade barriers, especially for developing nations which are not able to participate in global trade owing to cost considerations".

For example, smart contracts can be used for transactions to exchange money, property, shares, or anything of value where blockchain defines the rules and penalties of the contract and automatically enforce those obligations, including expiration dates and validity.

Additionally, blockchain can be integrated in the Internet of Things device management, voting confidence, and P2P insurance. All of these applications of blockchain are already being vetted and show favourable results.

We aim to address some of these big issues as a future job and build alternatives to them. We also intend to incorporate automated payments and delivery proof into our suggested solution-whereby parties are paid in a centralized and efficient manner using cryptocurrencies through smart chances for effective physical plant and product distribution.

The golden rule which states “those who have the gold, rule” seems probable as access to working capital will become necessary in a more transparent environment. Overall, “when adopted at scale, the solution has the potential to save the industry billions of dollars and benefit global trade and the world economy”.

CONCLUSION

Blockchain makes it possible for a system of independent actors to share and trust a record of digital assets, transactions, and information. In doing so, it offers the potential to disrupt and transform existing business models. When a blockchain-based traceability system for the full end-to-end supply chain for a given food commodity is not feasible, a minimum viable product should be prioritized.

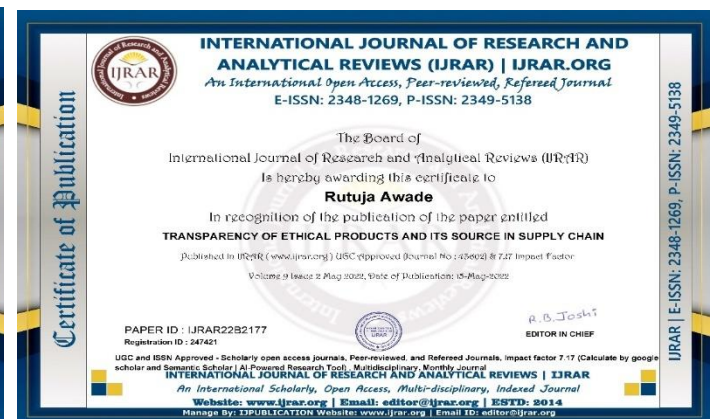
The successful application of blockchain requires strong planning, mobilization, and coordination. Private, public, and hybrid blockchain solutions each have unique strengths and weaknesses depending on specific requirements. It is not necessary to build applications on a public blockchain to reap the benefits of transparency and accountability. In addition to core business drivers and financial measures of success, blockchain has potential to drive impact in social good use cases, such as environmental sustainability

The important factor in supply chain management is their information and management. The information technology can help in this connection. With use of this technology, remarkable progresses do in all parts of supply chain. In addition to accomplished operations facilitating in supply chain, use of information technology can remove many of problems which exist in this connection. It is not necessary for all supply chain data to be accessible across all actors. Actors should agree on a data model that can share required traceability data and protect sensitive product data on the blockchain. Supply chain actors may need to invest in closing the digital divide to participate.

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CERTIFICATES



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Chapter 2

Chapter 3

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Chapter 7

Chapter 8